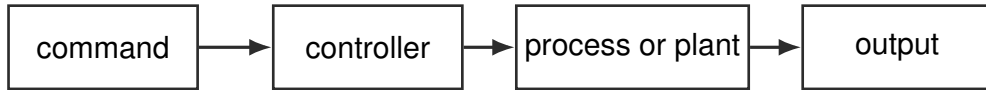
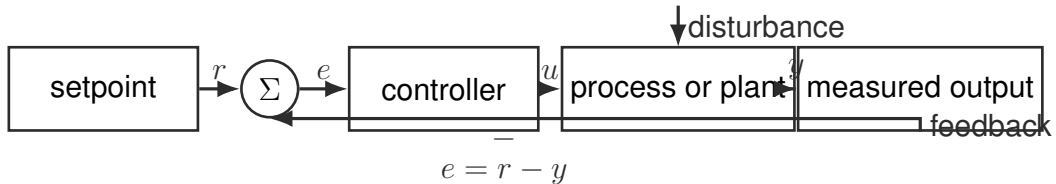


Open-loop control

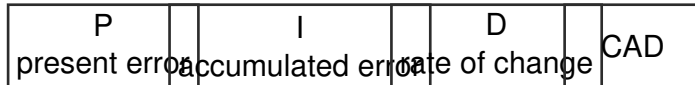


No measurement returns to correct the action

Closed-loop negative feedback



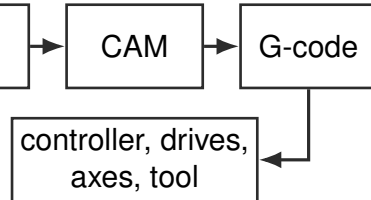
PID controller



$$u = K_P e + K_I E_{\text{sum}} + K_D \Delta e$$

No calculus is required for the classroom model.

CNC control chain



Home and limit switches provide machine-state evidence.